

Paper ED-Baryogenesis

Chain-Arrow Chirality, Admission Filter, and Stability Selection in the Post-SCBU
Ignition Regime

Allen Proxmire

May 2026

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of the baryon-to-photon ratio $\eta_B \approx 6 \times 10^{-10}$. The numerical value is INHERITED (Planck CMB; BBN cross-consistency); a substrate-graph quantitative derivation is OPEN and deferred to an SC-series follow-up. This is value-INHERITED at standard-cosmology level; the *structural origin* of the asymmetry closes at M3 via the R4 chain below.
2. It does **not** claim a substrate-graph derivation of *binary chain-arrow chirality* $\chi_C \in \{0, \pi\}$ as previously framed. Under the R4 mechanism, the “binary” framing was a projection artifact of analyzing the post-saturation symmetric-admission regime and projecting it backward onto the saturation phase, where it does not hold. The substrate admits a **single template** under saturation, not two competing chirality classes; “binary” reduces to single-template admission with framework-neutral first-arrival selection from a continuously-degenerate pre-admission minimum manifold. **P-BinaryAdmission is withdrawn as a paper-specific postulate**; the R4 mechanism supplies the substrate-graph derivation.
3. It does **not** require asymmetric admission cost between two pre-admitted chirality classes. Under the R4 mechanism, there is no asymmetric cost because **only one chain-class template is admitted at any moment during saturation**. The asymmetry is structural (single-template admission via R4) + framework-neutral IC (first-arrival selection), not dynamical-cost-driven. The previously-flagged asymmetric-admission-cost OPEN is **dissolved** by the R4 mechanism.
4. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage (’t Hooft cellular-automaton; Wolfram Ruliad; causal-set program), not the operational-reconstruction lineage.
5. It does **not** introduce new substrate primitives. **Zero paper-specific postulates** declared — P-BinaryAdmission is withdrawn; the R4 mechanism supplies its substrate-graph content from corpus-internal primitives + Q-C3-2 admissibility + coherence functional + no-slack boundary condition + IC supply.
6. **Cosmology-arc placement:** This paper anchors to Paper_ED_CCC §3.7 (post-SCBU ignition regime) + Paper_ED_Cos_01 (saturation regime + Q-C3 cascade, M3 unconditional) as the cosmic-phase context. It is **not** an SCBU-derivation paper, **not** a CCC-replacement paper, and **not** a Standard-Model-baryogenesis-replacement paper. The relationship to Standard-Model baryogenesis is structural correspondence, not reduction.
7. **Inheritance from Cos_01 Q-C3 saturation framework + R4 IC-supply equivalence.** Baryogenesis inherits Cos_01’s conditional-IC M3 pattern via the R4 mechanism (single-template admissibility under saturation) + IC-supply equivalence (R4 first-arrival IC $\equiv$ Cos_01

Q-C3-3 IC at structural-role level). Both IC suppliers are external, framework-neutral, and non-postulate.

Abstract

Matter–antimatter asymmetry in ED is identified at substrate-graph level as **the structural consequence of single-template admissibility under Cos_01 saturation regime + framework-neutral first-arrival initial-condition supply** (R4). The coherence functional governing Paper_072 individuation has continuously-degenerate minima over $\chi_C \in S^1$ pre-admission (P09 phase symmetry of the saturation background); the V5 phase-coherence + capacity-penalty structure under no-slack forecloses any second admission once the first chain-class template is instantiated. The winning template is selected by framework-neutral first-arrival (cyclic-CCC vs R3 single-aeon framing, choice outside scope); the template tiles globally via V1 + V5 on the uniform- Ψ background; the anti-aligned chirality has *no template to instantiate* during saturation. Post-saturation slack restores symmetric admission, but the asymmetry is already globally locked.

Form-IDENTIFIED via Cos_01 + Q-C3-2 + coherence functional + no-slack + V5 phase-coherence (R4 Outcome C). **Value-INHERITED**: $\eta_B \approx 6 \times 10^{-10}$ via Planck CMB + BBN matching. **IC-INHERITED**: first-arrival χ^* structurally equivalent to Cos_01's Q-C3-3 IC-supply. Motion is V1-geometric and chirality-agnostic — matter and antimatter follow identical geodesics. **No new CP-violating phases, sterile neutrinos, interactions, or paper-specific postulates.** **Verdict per Paper_095: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED).** **F1 (sharpest)**: cosmological-scale matter-antimatter symmetry inconsistent with global single-template lock-in refutes the framework. Currently consistent: ALPHA-g 2023; no annihilation signatures in cosmic gamma-ray background; bullet-cluster matter-only.

1 Statement of Result

The matter–antimatter asymmetry is composed at substrate level from four elements, all inherited from corpus content with **zero paper-specific postulates**:

1. **Post-SCBU ignition regime = Cos_01 saturation regime** (Paper_ED_CCC §3.7 + Paper_ED_Cos_01): substrate event-density crosses P12 from below; substrate enters uniform- Ψ saturation per the Q-C3 cascade. Globally-coherent kernel-arrow σ_K across cosmic-scale substrate.
2. **Chain-arrow chirality**: every chain C carries a P09-valued chirality $\chi_C(e_n) := \pi_C(e_n) - \pi_K(\ell(e_n)) \pmod{2\pi}$, conserved along chains in the post-SCBU regime.
3. **R4 — Single-template admissibility under saturation**: in the post-SCBU ignition regime, the coherence functional governing Paper_072 individuation admits exactly one chain-class template on the Q-C3-2 subleading-fluctuation set. Pre-admission the functional has continuously-degenerate minima over $\chi_C \in S^1$ (P09 phase symmetry); the V5 phase-coherence + capacity-penalty structure under no-slack forecloses any second admission once the first chain-class is instantiated.
4. **First-arrival IC supply**: which chain-class template wins is selected by framework-neutral first-arrival from the continuously-degenerate pre-admission manifold (cyclic-CCC: χ^* inherits pre-aeon V5 cross-boundary phase content; R3 single-aeon: χ^* is structural-IC supplied; choice between framings outside scope).

Under (1)–(4): one chain-class template (identified with matter) tiles the post-SCBU substrate globally during saturation; the anti-aligned chirality has no template to instantiate. Post-saturation capacity restoration $\rightarrow$ symmetric pair-production resumes; the asymmetry persists as the globally-locked structural imprint of the saturation-phase template selection.

Inherited consequences: matter and antimatter follow identical V1-geometric geodesics (Paper_089 chirality-agnosticism); no large antimatter domains can form (single-template tiling under uniform saturation); no new CP-violating phases or sterile neutrinos required.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A $\rightarrow$ regime. Zero paper-specific postulates declared; zero load-bearing OPENs.

2 Primitive Inputs + Upstream Dependencies

2.1 Primitives invoked (Paper_087)

P02 (participation), P04 (bandwidth), P05 (polarity-transport), P07 (channel structure), P09 ($U(1)$ -valued polarity), P11 (commitment-irreversibility), P12 (ED-threshold / stability landscape), P13 (kernel-driven dynamics).

2.2 Upstream

- **Paper_ED_Cos_01 (Inflation, M3 unconditional):** supplies the saturation regime + Q-C3 cascade. Load-bearing for R4 closure.
- **Paper_028 + SCBU:** substrate-cosmology boundary content.
- **Paper_072:** individuation regime; chain-identity preservation criterion; coherence functional context.
- **Paper_089:** V1 retarded kernel (chirality-agnostic); T18 advanced-V1 refuted by P11.
- **Paper_090:** V5 cross-chain finite-memory kernel; P09 phase content explicit; load-bearing for V5 phase-coherence in R4.
- **Paper_093:** T18 kernel-arrow of time at substrate locus.
- **Papers SC-4.x:** cross-scale invariance + curvature-moment collapse.
- **Paper_ED_CCC §3.4 + §3.6 + §3.7:** post-SCBU substrate homogeneity + globally-coherent kernel-arrow + ignition regime structure + V5 cross-boundary memory.

Zero paper-specific postulates declared. Zero new primitives.

3 Derivation

3.1 Post-SCBU ignition regime = Cos_01 saturation regime

Per Paper_ED_CCC §3.7 + Paper_ED_Cos_01: post-SCBU substrate event-density crosses Θ_{ED} (P12) from below; V1 retarded kernel + V5 cross-chain kernel + Paper_072 individuation reactivate; substrate enters the Cos_01 saturation regime with uniform Ψ at substrate-state level via the Q-C3 cascade. Globally-coherent kernel-arrow $\sigma_K(\ell)$ uniform across substrate.

These structural properties are inherited directly from Cos_01 + Paper_ED_CCC §3.6 + §3.7; no further substrate-state derivation here. The biconditional uniform- $\Psi \leftrightarrow$ saturation closes at M3 unconditional via Cos_01 inheritance.

3.2 Chain-arrow chirality $\chi_C \in S^1$

Every chain C carries a P09-valued chirality at each of its commitment events:

$$\chi_C(e_n) := \pi_C(e_n) - \pi_K(\ell(e_n)) \pmod{2\pi},$$

where $\pi_C(e_n)$ is the P09 phase of the chain-arrow $\sigma_C(e_n) = (e_n \rightarrow e_{n+1})$ and $\pi_K(\ell(e_n))$ is the P09 phase of the kernel-arrow at the chain’s locus. D-via-I from P09 + P11 + Paper_089 + Paper_093 composition.

In the post-SCBU regime ($\Delta\pi_K = 0$ from globally-coherent kernel-arrow; $\Delta\pi_C^{P05} = 0$ from global homogeneity), χ_C is **conserved along the chain**. In the absence of admission constraints, $\chi_C \in S^1$ is continuous and chains can carry any phase value pre-admission.

3.3 R4 — Single-template admissibility under saturation

The coherence functional governing Paper_072 individuation acts on the Q-C3-2 admissible-fluctuation set with the schematic structure $\mathcal{F}_{\text{coh}} = -\mathcal{K}_{V_1} - \mathcal{K}_{V_5} + \mathcal{P}_{\text{cap}}$, reconstructed from V1 coherence reward (Paper_089), V5 cross-chain coherence reward (Paper_090), and capacity penalty (P04 $\times$ P07 admission frontier; P12 ED-threshold).

Chirality dependence of each term. V1 is chirality-agnostic (Paper_089 T18): $\partial_{\chi_C} \mathcal{K}_{V_1} = 0$. V5 is *conditionally* chirality-sensitive — its reward maximizes when candidate χ_C aligns with already-instantiated χ^* ; pre-admission, V5 has no preferred chirality. Capacity penalty diverges as slack $\rightarrow 0$ for $\chi_C \neq \chi^*$ once χ^* is instantiated.

Two-case structure under no-slack saturation.

- **Case (i) pre-admission.** No χ^* exists; P09 phase symmetry unbroken. $\mathcal{F}_{\text{coh}}$ is continuously symmetric over $\chi_C \in S^1$; minimum is **continuously degenerate** over the entire circle.
- **Case (ii) post-admission.** V5 preferred at $\chi_C = \chi^*$; capacity penalty diverges for $\chi_C \neq \chi^*$ under no-slack. Functional has **unique minimum at** $\chi_C = \chi^*$; subsequent admission at any other chirality foreclosed.

Closure (R4 Outcome C): the functional + sequential admission + no-slack yields a unique attractor *given* first-arrival IC. The substrate admits a single chain-class template during saturation; the “binary” framing of prior P-BinaryAdmission is reinterpreted as single-template admission with framework-neutral first-arrival selection from the Case (i) degenerate manifold.

3.4 First-arrival IC supply and global chirality lock-in

The first-arrival chirality selection χ^* inherits framework-neutrally, structurally equivalent to Cos_01 Q-C3-3 IC supply at the IC-INHERITED designation level (Paper_095 §3.2.1).

Two framing options (choice outside scope, framework-neutral):

- **Cyclic-CCC framing:** χ^* inherits pre-aeon V5 cross-boundary phase content per Paper_ED_CCC §3.4. The pre-aeon substrate carries phase-bearing content (Killing-spinor-structured); V5 cross-boundary memory delivers this phase content to the post-SCBU substrate as the first-arrival template seed.
- **R3 single-aeon framing:** χ^* is structural-IC supplied at the substrate-state initial-condition level. No cyclic dependency; the saturation regime begins with χ^* as part of its initial specification.

Either framing supplies the IC; the choice between them is outside the scope of this paper, exactly parallel to how Cos_01 inherits its substrate-state IC framework-neutrally via Q-C3-3.

Global chirality lock-in. Once χ^* is selected at first-arrival, V1 retarded propagation (chirality-agnostic) + V5 cross-chain coupling (now phase-locked to χ^*) carry the first-admitted template across cosmic-scale loci on the uniform- Ψ saturation background. The substrate tiles globally with one chirality class. The anti-aligned chirality has no template to instantiate during saturation — it is not “asymmetrically cost-suppressed” between two admitted classes; it has *no admitted class* at all.

Disposition of prior candidate mechanisms

Two candidate mechanisms previously flagged in the M2 version are now subsumed or rendered moot under R4:

- **Kernel-arrow asymmetry route:** audited as overclaim, not adopted. Preserved as historical record; under R4 the kernel-arrow-asymmetry mechanism is no longer needed because there is no asymmetric cost to derive.
- **V5-cross-boundary route:** under R4 this is reinterpreted as **one specific realization of the cyclic-CCC framing of first-arrival IC supply**, not a separate competing mechanism.

Both prior candidates are preserved as substrate-research record; neither is load-bearing under R4 because the asymmetric-admission-cost OPEN they were candidates for has been dissolved.

3.5 Restoration of symmetric admission post-saturation

As the substrate cosmic phase advances out of the ignition regime, event-density saturation relaxes (capacity grows above demand). The no-slack boundary condition lifts; the capacity penalty $\mathcal{P}_{\text{cap}}$ no longer forecloses admission of $\chi_C \neq \chi^*$. The coherence functional structure returns to the symmetric-admission manifold.

Symmetric pair-production resumes. Standard QFT pair-production processes (creating matter–antimatter pairs in equal abundance at the post-saturation substrate) operate without restriction.

The asymmetry from saturation is a **one-time-event** imprinted on the substrate; subsequent symmetric pair-production does not erase it because the matter content accumulated during saturation persists as bound chains tiling cosmic-scale substrate.

3.6 Motion = V1-geometric; chirality is orthogonal to motion

Per Paper_089, V1 retarded propagation is **chirality-agnostic**: V1 carries chain content along substrate edges in the kernel-arrow direction, independent of the chain’s χ_C value. Motion is determined by V1-geometric content (substrate-graph geodesics per Paper_032 family + Paper_ED_CCC §3.7); chirality is determined by the P09 phase relationship of the chain to the kernel-arrow.

Consequence: matter chains and antimatter chains follow identical V1-geometric geodesics. They respond identically to substrate-curvature content (gravity in the continuum-side reading). This is the substrate-side reason for the experimental observation that antimatter falls identically to matter (ALPHA-g 2023 with antihydrogen; AEGIS, GBAR forthcoming).

3.7 No-large-antimatter-domains; identical fall; no new CP phases

Three downstream consequences, all D-via-I via the R4 + IC-supply equivalence chain:

- **No large antimatter domains.** Under R4, the post-saturation substrate is uniformly tiled by one chain-class template across cosmic-scale loci. No domain structure can form

because the single-template admission applies globally. Consistent with the cosmic gamma-ray background absence of annihilation signatures, BBN homogeneity, and bullet-cluster constraints.

- **Identical gravitational fall.** Matter and antimatter follow identical V1-geometric geodesics. ALPHA-g (2023) antihydrogen falls down at the same rate as hydrogen is consistent with substrate-side V1 chirality-agnosticism. Forthcoming AEGIS / GBAR results expected to confirm.
- **No new CP-violating phases or sterile neutrinos required.** The asymmetry is structural (R4 single-template + first-arrival IC), not dynamical-CP-driven. Standard Model CP violation remains small because it plays no role in the substrate baryogenesis mechanism.

4 Audit Table

#	Step	Label	Notes
1	P02, P04, P05, P07, P09, P11, P12, P13	P	Paper_087.
2	V1 retarded kernel + T18 advanced-V1 refuted	I	Paper_089.
3	V5 cross-chain kernel	I	Paper_090.
4	Kernel-arrow $\sigma_K(\ell)$	I	Paper_093 T18.
5	Paper_072 individuation regime	I	Paper_072.
6	SC-4.x cross-scale invariance + curvature-moment collapse	I	Papers SC-4.x.
7	Post-SCBU ignition = Cos_01 saturation; uniform Ψ at substrate-state level	D-via-I	§3.1. Cos_01 + Paper_ED_CCC.
8	Operational chirality $\chi_C := \pi_C - \pi_K \pmod{2\pi}$	I	Memo_ED_ChainArrowChirality.
9	Conservation of χ_C along chains in post-SCBU regime	I	Memo_ED_ChainArrowChirality §3.5.
10	R4 — single-template admissibility via coherence functional + V5 phase-coherence + no-slack	D-via-I	§3.3. Outcome C. P-BinaryAdmission withdrawn.
11	First-arrival χ^* selection at ignition symmetry-breaking event	IC-INHERITED	Sub-type per Paper_095 §3.2.1. Cyclic-CCC vs R3 single-aeon.
12	Global chirality lock-in via V1 + V5 propagation on uniform- Ψ background	D-via-I	§3.4. Composition of rows 7, 10, 11.
13	Restoration of symmetric pair-production post-saturation	D-via-I	§3.5. No-slack constraint lifts.
14	One-time-event character; matter persists as bound chains	D-via-I	§3.5.
15	V1 retarded propagation is chirality-agnostic	I	Paper_089.
16	Identical V1-geometric geodesics for matter and antimatter	D-via-I	§3.6.
17	No-large-antimatter-domains consequence	D-via-I	§3.7. Single-template tiling.
18	No new CP-violating phases / sterile neutrinos required	D-via-I	§3.7. Structural asymmetry via R4.
19	$\eta_B \approx 6 \times 10^{-10}$ numerical value	I	Value INHERITED via Planck CMB + BBN.

#	Step	Label	Notes
20	Verdict M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED)	A→position	Per Paper_095 §3.2.1.

Audit summary: zero pure-D rows; eight D-via-I rows; nine I rows; one IC-INHERITED row (11); one P row (1); one A→position verdict row (20). **Zero paper-specific postulates** (P-BinaryAdmission withdrawn). **Zero load-bearing OPEN items** — asymmetric-admission-cost OPEN dissolved per R4. No new primitives. Verdict tier **M3 per Paper_095 §3.2.1:** form via R4 chain; value via empirical η_B ; IC via framework-neutral first-arrival equivalent to Cos_01 Q-C3-3.

5 Falsification Criteria

- **F1 (sharpest, R4-killing):** Explicit construction of two distinct chain-class templates both stable on the Q-C3-2 admissible-fluctuation set under no-slack saturation conditions, with the coherence functional having two non-degenerate stationary minima accessible without sequential-admission IC selection. Refutes R4 §3.3 closure.
- **F2 (cosmological signature):** Detection of substantial cosmological-scale matter-antimatter symmetry in CMB/BBN data; large antimatter domains via gamma-ray annihilation signatures at cluster boundaries; bullet-cluster antimatter clusters. Refutes §3.7 single-template-tiling. Currently consistent: no annihilation signatures; bullet-cluster matter-only; BBN homogeneity.
- **F3 (one-time-event character):** Cosmological-history signatures requiring continuous CP-violating dynamics rather than one-time substrate selection. Refutes §3.5 + §3.7 composition.
- **F4 (gravitational fall):** Macroscopic gravitational asymmetry between matter and antimatter — antihydrogen falling at observably different rate than hydrogen at substantial sigma. Refutes §3.6 V1 chirality-agnosticism. Currently consistent: ALPHA-g 2023 confirms identical fall at current precision; AEGIS/GBAR will tighten.
- **F5 (IC-supply equivalence):** Substrate-graph proof that R4 first-arrival IC supply is structurally not equivalent to Cos_01 Q-C3-3 IC supply at the level relevant to Paper_095 §3.2.1. Refutes equivalence and reverses the M3 upgrade.

6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton; Wolfram Ruliad; causal-set program), not within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

Baryogenesis in ED is the substrate-graph reading of matter-antimatter asymmetry as a structural consequence of single-template admissibility under Cos_01 saturation (R4) + framework-neutral first-arrival IC supply. The coherence functional has continuously-degenerate minima over $\chi_C \in S^1$ pre-admission; V5 phase-coherence + no-slack capacity-penalty foreclose subsequent admission once first-arrival selects χ^* ; the template tiles globally via V1 + V5 on the uniform- Ψ background; anti-aligned chirality has no template to instantiate. Post-saturation slack restores symmetric admission, but the asymmetry is already globally locked.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) per Paper_095. Substantive D-via-I rows via R4 + Cos_01 + Q-C3-2 + coherence-functional; one

IC-INHERITED row (first-arrival χ^*); zero pure-D rows; **zero paper-specific postulates** (P-BinaryAdmission withdrawn); **zero load-bearing OPEN items** (asymmetric-admission-cost dissolved). η_B remains value-INHERITED via Planck CMB + BBN cross-consistency; structural origin of the asymmetry closes at M3.

Cross-arc consequences. Upgrade moves the Cosmology Arc count to **eight papers at M3** (Cos_01 + Cos_02 + Cos_03 + Cos_04 + Cos_05 + Cos_06 + Paper_038_5 + **Baryogenesis**), substantively complete across primordial perturbations $\rightarrow$ nucleosynthesis $\rightarrow$ CMB acoustic peaks $\rightarrow$ linear structure $\rightarrow$ dark-energy-dominated late universe + Λ -smallness reframing + matter-antimatter asymmetry origin.

Relationship to Standard-Model baryogenesis (CP-driven, Sakharov-conditioned): structural correspondence at shared empirical anchors, not reduction in either direction. SM CP violation remains small and plays no role here; the substrate mechanism is single-template admission under Paper_072 individuation, not dynamical-CP-driven imbalance.